

Neel Modha

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SKILLS

Programming Languages: C++, Python, HTML/CSS, TypeScript, MATLAB

Tools: Visual Studio Code, Git/GitHub, SolidWorks, Raspberry Pi, ESP32/Arduino, GitHub Pages

EXPERIENCE

Automation Engineering Intern, ESCO Group

May 2026 – Aug 2026 Marion, IA

- Programmed programmable logic controllers (PLCs) using ladder logic for industrial automation and control systems.

Material Handler III, Pearson

Dec 2025 – Present Marion, IA

- Organized inventory and maintained accurate counts across warehouse locations.
- Palletized boxes efficiently for shipment preparation and warehouse organization.
- Operated an electric forklift safely to transport materials throughout the facility.
- Ensured workplace safety compliance and maintained a clean, organized warehouse environment.

Assistant Tennis Coach, Bowman Woods Racquet Center

Jun 2024 – Aug 2024 Marion, IA

- Coached tennis to players ages 10–18, developing their skills through structured practice sessions.
- Ran multiple sessions daily and assisted with training drills to improve technique and performance.
- Collaborated with the head coach to develop progressive training programs for student athletes.

Sales Associate, Dollar Tree

Jun 2022 – Jul 2023 Marion, IA

- Assisted customers with product selection and provided excellent customer service.
- Managed cash register operations and maintained accurate transaction records.
- Restocked shelves and organized store displays to ensure a clean, accessible shopping environment.

PROJECTS

ESCO Automation Studio — Offline-First PLC Engineering IDE

TypeScript, React

- Built a Studio 5000–style ladder logic editor that stores rungs as a semantic AST rather than text, supporting nested parallel branches, project-wide undo/redo, and rung copy/paste.
- Implemented **.L5X** import/export with line-by-line round-trip preservation and a Verify pass that flags undefined tag references, array bounds, and CASE selector type mismatches.
- Wrote an offline scan-cycle simulator executing a Structured Text subset (IF/ELSIF, bounded FOR/WHILE/REPEAT, JSR/SBR/RET) plus a read-only EtherNet/IP driver for ControlLogix and CompactLogix tag access.

ESCO Engineering City — Engineering Review & Delivery Platform

TypeScript, Three.js, Python

- Built an interactive 3D city interface of 23 operational zones that visualizes live project state — agent locations, task assignments, and event replay — with orbit, zoom, and first-person walk modes.
- Implemented source intake across L5X, CSV, XLSX, DOCX, PDF, and XML with exact evidence anchors, feeding a durable resumable task graph that extracts requirements and prepares release candidates.
- Shipped a Release Center producing immutable release archives with SHA-256 manifests behind six named human approval gates and a full audit trail.

ASME IAM3D Mars Rover (R.O.V.E.R.) — Team Lead, Software/Electrical

Aug 2025 – Present

- Lead a 10-member engineering team through the complete design, development, and testing of an autonomous Mars rover for the ASME IAM3D national competition.
- Own all software and embedded systems development including autonomous navigation, robotic arm control, and telemetry, using Python, C++, and TypeScript.
- Ensure safety compliance, coordinate collaboration between mechanical and electrical subsystems, and manage project timeline and deliverables.

Personal Portfolio Website

TypeScript, WebGL2

- Designed and built a portfolio site whose signature element is a from-scratch WebGL2 Navier–Stokes fluid solver rendering ink on paper, with a graceful fallback when float render targets are unavailable.
- Deployed through a GitHub Actions pipeline to GitHub Pages.

EDUCATION

University of Iowa — BSE Computer Science & Engineering

Expected May 2029

GPA: 3.5

Relevant Coursework: Intro to Engineering Computing, Intro to Engineering Problem Solving, Computer Science 1, Matrix Algebra, Discrete Structures, Physics, Chemistry